

Tackling Discrimination in Tech: The Anti-Bias Cards (ABC) in Use

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ABSTRACT

How to engage with the consequences of systemic biases and subsequent calls to ensure more equity in technologically oriented teams without any prior education on these matters presents an open question within Human-Computer Interaction (HCI). Based on a previous literature study and the development of the Anti-Bias Card Deck (ABC), this paper presents the findings and insights from various workshops in which the cards were applied. The settings varied in their thematic and geographic contexts as well as with regard to participants and their backgrounds. Overall, participants agreed that the card deck could be of help to detect and tackle different aspects of discrimination over a project's life cycle. Hence, the ABC deck fulfilled its intended role as a potential conversation starter for reflecting on matters of bias and equity in teams of technology developers without prior expertise.

CCS CONCEPTS

• **Social and professional topics** → **Gender**; • **Human-centered computing** → *HCI theory, concepts and models*; Empirical studies in HCI.

KEYWORDS

gender, sensitivity, methodology, intersectionality

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1 INTRODUCTION

Reflecting on how projects in technological contexts are affected by systemic aspects of inequality and discrimination is becoming more and more important to researchers and practitioners when planning, seeking funding for, executing and reporting on (or marketing) their projects [1]. For example, many funding bodies ask for how project goals attend to social issues, or how inequality is handled within projects themselves¹. At the same time, researchers

¹For instance, the European Commission requires Horizon Europe applicants to integrate “the gender dimension”: <https://ec.europa.eu/info/funding-tenders/>

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and practitioners develop their sensitivity about societal inequality, and might want to make sure that they do not perpetuate injustices (c.f. [5]). However, without any deeper background understanding of how these issues come to be, it may be very challenging to identify and answer relevant questions. Hence, we developed the Anti-Bias Card deck (ABC) as a tool for teams to ask and discuss relevant questions to uncover internalized, unintentionally accepted, and unacknowledged mechanisms of oppression and exclusion – and, thus, to take a first step toward tackling these issues.

After completing previous work on the card deck [2], the first author conducted five case studies where the ABC was used during the last two years. These cases span across different target groups, group sizes, and slight differences in geographic setting.

2 A LOOK BACK: WHAT IS THE ABC?

The first author was working as an intern at a research institution in Vienna, when her superior asked for help with a funding application. The funding organization was asking for an explanation on how the project they were working on would tackle gender inequality, either in general, or within the context of their field (computer vision). The project itself had no obvious gender aspects, so this question was difficult for the team to answer – but also forced them to ask it in the first place. In the following years, in conversations with researchers from other fields and areas, the first author realized that inconspicuous influences and unexpected outcomes appear to be comparatively hard to pin down when you do not know what you could or should be looking for, even (or especially) if the context is something that you have been looking at for quite a while (e.g. a project that's been running for a bit). Hence, we searched to offer a starting point to students, researchers, and practitioners to explore questions and issues around gender and other factors of discrimination, and how these relate to their research and practice.

In this context, we created a set of recommendations [1], which were then put into the shape of a card deck. We decided to create a card deck because we consider the topic to be too complex and content-dependent for e.g. a checklist. Above all, we intended the card deck to prompt reflection, which is why it asks questions, provides explanations, and offers ideas for suitable measures, comparable to e.g. the HCI Guidelines for Gender Equity and Inclusivity [8]. Additionally, it is usable in different settings, from single users, via small groups to large groups.

2.1 Recommendations

We describe the process of developing the recommendations elsewhere [1]. In short, we conducted a close reading [6] of selected

opportunities/docs/2021-2027/horizon/guidance/programme-guide_horizon_en.pdf, accessed 2023-03-04.

papers published to the ACM digital library and partly selected from [9], where we went looking for pointers towards (gender) sensitivity in HCI. From these works, we extrapolated some practical suggestions and grouped them according to project life cycles: *designing* research, *acquiring funding*, *conducting* research, and *presenting* research. From our engagement and ensuing discussions, we derived a set of high level recommendations on considering different markers of identity (with a focus on gender) during various stages of a project life cycle.

2.1.1 Designing Research Protocols. The cards for this phase include questions and ideas to help us *Articulate Gender Explicitly*, e.g. "What do we mean when we talk about gender?". To account for the influence our position can have on our work, there are recommendations to *Plan for Individual and Collective Reflection on Positionality*. As different methods work (not so) well for different people, we advise to *Probe Methods for Accessibility and Sensitivity*. And finally, when we have set up our plans, we should reflect on them, and *Deliberately Assess Exclusions*.

2.1.2 Acquiring Funding. Organizations providing funding often operate with a traditional, binary view on gender, or not much awareness of discrimination at all - thus, we propose to *Understand Proposal Writing as a Teaching Opportunity*. Additionally, with limited understandings of oppression in those organizations, it might be interesting to *Check Expectations and See How They May Be Extended*.

2.1.3 Conducting Research. For this phase of a project, the cards offer advice on how to *Follow Participants' Choices In Identifying Them* and to *Attend to Different Needs and Preferences*. To find gaps and missteps early on, they include recommendations to *Seek Critical Feedback* as well as to *Actively Look for What's Missing*.

2.1.4 Presenting Research. Talking about our work can happen at any point in time, of course. In any case, it will be important to *Provide Appropriate Context Information* to the readers. The cards also offer ideas and pointers to *Reflect on Representation and Prior Work* and to *Choose Mindful Language*.

2.2 Card Design

In this section, we briefly look back at our intentions and ideas when creating the card deck, and when designing the cards in both physical and digital modalities. We start out with the general question, "Why even create a card deck?", and then go into details as to the layout of the cards.

As discussed in [2], we wanted to provide HCI researchers and practitioners with a flexible tool that serves to start a process of reflection, and be applied in different group settings. Individuals can use the card deck by themselves, and there are several different ways for groups to engage with the material. Rather than looking at and checking off yet another static set of comparatively abstract recommendations, a project team may sit down together and try to answer the prompt questions and discuss the concepts presented on the cards. For example, a group of volunteers trying to make their repair café more accessible to marginalized groups could use the deck to assess how they could include and acknowledge various dimensions of marginalization in their structures. We assumed that

the haptic nature of a card deck – something people could be toying with while thinking and discussing about its contents – would offer an additional "hands-on" aspect to the process. With the card deck, we intend to offer researchers and practitioners a way to interact and engage with a range of different perspectives to investigate their project from. When used in a group setting and, e.g., handing out one card per participant, each participant will have to assume a specific position, different from the positions of others, from which to discuss the project in question.

As can be seen in Figure 1, each card consists of some prompt questions, a short text providing general information and proposing actions to take, and (optionally) footnotes with explanations. To increase the accessibility of the content, the cards are written in plain German and plain English. This means the first author in an act of translation intentionally removed or explained technical and academic terms common within Gender Studies and HCI, to not put up additional barriers to the content. Terms that are used on multiple cards have explanatory footnotes on each, so that the cards can be used on their own. This was done to avoid creating a power and knowledge imbalance by having one participant have the control over, e.g., a glossary or list of terms.

2.3 Online card deck

The initial plan for testing the card deck was to use it in a physical setting with researchers at TU Wien. However, as an opportunity to test the card deck in an online setting became more feasible due to the Covid-19 pandemic, the online version became high priority. Inspired by various online tools that let users "draw" random items from selections (e.g. the maurudor website which would upon reload present the user with a random cat picture²), the web version started out as a website serving screenshots of the cards³. An example for the current layout (see also <https://div.uber.space>) is presented in Figure 1b. The only difference in comparison to the physical cards' layout are the buttons to draw a new card in either German or English (top right, dice symbols) and the link back to the description of the tool (top right, "i" symbol).

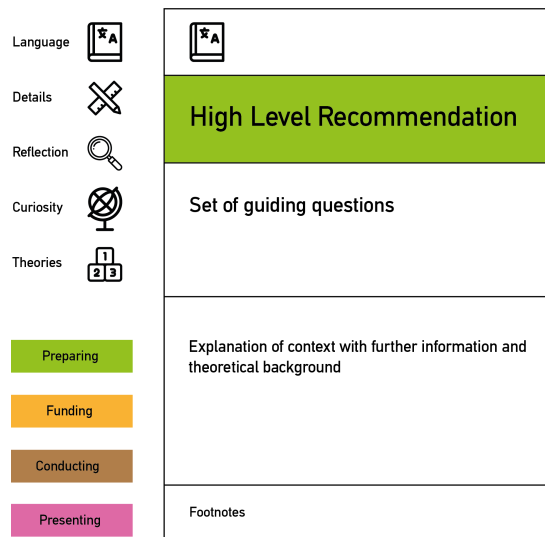
3 CASE STUDIES' SETTINGS

In this section, we provide an overview of how the workshops using the card deck took place: the context, the methods used, and the participants. The entries in the section are sorted chronologically. Each subsection consists of a short description of the case study context and tools used for communication, presentation, and gathering of feedback. A short overview is given in Table 1. We summarize and discuss the feedback provided by all participants collectively in Section 4.

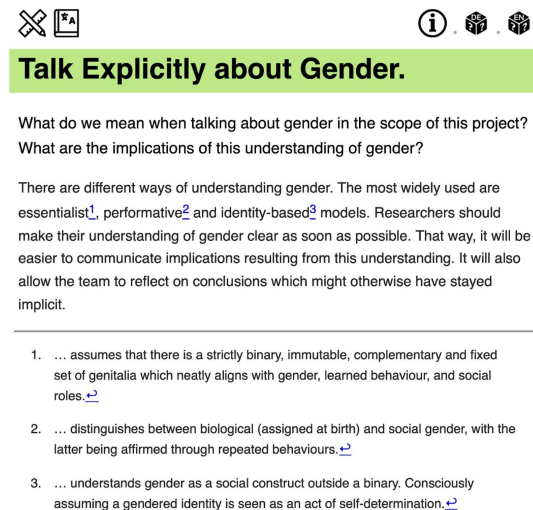
Procedures were comparable in four of five case studies (the exception being Ways of Thinking, described in Section 3.3): After introducing herself, the first author asked participants in their introductions to talk about their current connection to diversity and inclusion. She further inquired whether they had any current projects they would like to apply the card deck's recommendations to (this request was included prior in the workshop descriptions).

²<https://web.archive.org/web/2023000000000000/maurudor.de>, accessed 2022-03-04

³The first author is fully aware of how ironic it is that screen reader and mobile accessibility were only an afterthought of the online version, and is thankful to the people who helped out with fixing this(see [2]).



(a) Card schematic showing structuring elements (icons, color) and text composition [2].



(b) Screenshot of a card from the online version, produced 2023-01-03.

Figure 1: Card design examples (general and specific).

After introductions, the group would briefly discuss their individual experiences with diversity and inclusion measures so far.

Afterwards, the first author followed up with a presentation and discussion of the development and content of the recommendations and card deck. In the case of online workshops, she would explain the functionality of the online card deck and drawing stack. Independent of online or offline setting, the first author offered space and time for questions after these introductions of people and tools.

3.1 ditact women's IT studies

In September 2020, the first author facilitated a half-day workshop presenting the recommendations and using the card deck at ditact

women's summer IT studies⁴. The goal of the workshop, for the authors, was to evaluate whether the card deck would be useful in an interactive setting, encouraging discussion and reflection on (gender) sensitivity in (HCI) research and practice. The workshop description is still available online (German only)⁵.

Five participants with different backgrounds attended the workshop. For them to engage with the cards, the first author designed an activity consisting of two parts: first, each participant was asked to draw a card for themselves and (by themselves) think of ways of applying its content to their project or occupation in general. After about ten minutes, the group met again in a plenary setting, and discussed the participants' findings together. As this workshop happened remotely, ditact provided a Zoom room. Documentation of the workshop happened in an online collaborative rich-text document. The main language used in the workshop and its documentation was German.

3.2 Gender Equality Academy

In April 2021, the first author conducted a half-day workshop as part of the H2020 project Gender Equality Academy⁶ (GE Academy). The project aims to develop and implement "a high-quality capacity-building programme on gender equality in research, innovation and higher education"[4]. The description of the workshop can be found online⁷.

Eleven people had signed up for the workshop, and five filled in the feedback questionnaire. The presentation of the card deck and recommendations has been recorded and is available on youtube⁸. As this group was bigger, engagement with the card deck happened in a slightly different setting as compared to the ditact workshop. Groups of two to three people each worked together on a Miro board. Their task was to try and apply two cards per group to their respective projects and fields of expertise. The groups were put into virtual breakout rooms for about 40 minutes. The Miro board had been prepared in advance by the GE Academy team, and populated with labeled group spaces, each containing card depictions similar to the cards in the online tool - see Figure 2.

This workshop, too, happened via Zoom. Group work discussions, ideas and findings were documented on the Miro board for discussion in the plenary setting afterwards. Like in the ditact workshop, the group returned to a plenary session to discuss what had been put on the board. Post-workshop feedback was gathered using a Google Form set up by the GE Academy team.

3.3 Ways of Thinking in Informatics

This case study diverges from the others, as it was part of a compulsory first-semester course for computer science students at TU Wien, and students could complete their tasks over the course of several weeks in the winter semester of 2021/2022.

⁴<https://ditact.ac.at/>

⁵<https://ditact.ac.at/course/1956-kurs-online-projekte-inklusiv-gestalten>, last accessed 2023-03-04

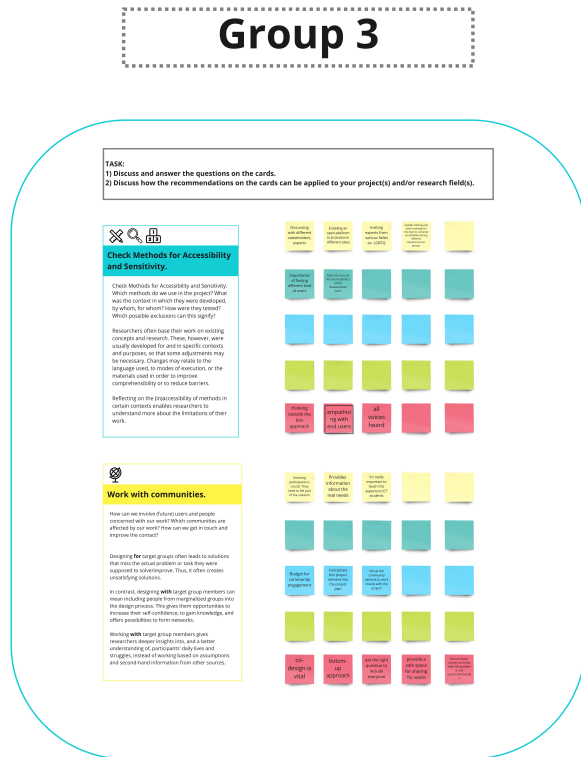
⁶<https://ge-academy.eu/>

⁷<https://ge-academy.eu/gender-sensitivity-in-human-computer-interaction-research/>, last accessed 2023-03-04

⁸<https://www.youtube.com/watch?v=klyXjBRsIo>, last accessed 2023-03-04

Table 1: Overview of Case Studies

Date	Case Study Name	Target Group	Setting	Tools used
September 2020	ditact women's IT studies	women students & workers in IT	online	Zoom, CryptPad
April 2021	Gender Equality Academy	researchers, team leads, coordinators etc. in universities & private companies	online	Zoom, Miro, Google Forms
Fall 2021/2022	Ways of Thinking in Informatics	first semester Computer Science (CS) students	online	Zoom, course platform
July 2022	May Contain Hackers (MCH)	people from the worldwide hacker community	offline	card deck
October 2022	Bits und Bäume 2022	activists & academics at the intersections of sustainability and digitalisation	offline	card deck

**Figure 2: Miro board after being filled in by one of the workshop groups at the GE Academy Workshop. Illustrative purpose only.**

Ways of Thinking in Informatics (“Denkweisen der Informatik” in German, or “Denkweisen” for short⁹) aims to teach students to

⁹<https://tiss.tuwien.ac.at/course/courseDetails.xhtml?dswid=8550&dsrid=3&courseNr=187B12>, last accessed 2023-03-04

“reflect critically on the contents and priorities in informatics” [3, 7]. The course consists of several chapters, each representing one “way of thinking”. To pass the course students have to complete “challenges” and show their understanding of the contents and concepts taught relating to the chapters. As theoretical input, the students received a presentation and discussion of discrimination issues in informatics as well as different approaches to mitigate them, including the recommendations presented in the card deck [1]. The students then were tasked to familiarize themselves with the recommendations, to read one of a selection of academic papers (each not published earlier than five years before the case study took place), and to critically reflect upon the selected paper using the recommendations.

All sessions related to this case study were conducted on Zoom, including the Q&A sessions the first author offered for the students to discuss open questions and uncertainties. Submission of the students’ work was done via the courses’ own online platform, which was additionally used to evaluate their work, and to gather feedback from the students regarding both the card deck and the challenge.

3.4 May Contain Hackers

In July 2022, the first author used the card deck in a face-to-face setting for the first time: she conducted a 90-minutes workshop at May Contain Hackers¹⁰, a Dutch hacker camp. The announcement for the workshop can be found online¹¹. Eight people participated in the workshop; most of them worked in informatics-related areas: software development, security research and education. One participant had already retired from his day job, and now worked as an usher in a theater.

Here, the first author wanted to see whether participants could use the cards as discussion starters or inspiration for whatever everyone brought with them - including other fields of research, work, or even activism. After a short round of introductions, and everyone drawing a card, the participants spent about ten minutes

¹⁰<https://mch2022.org/>

¹¹<https://program.mch2022.org/mch2022/speaker/QZT9AU/>, last accessed 2023-03-04

thinking them through on our own. After that, the group as a whole discussed each participants' cards, thoughts, and findings in the plenary.

3.5 Bits und Bäume 2022

Another face-to-face setting, the first author's workshop at Bits und Bäume (BuB)¹² took place on October 2, 2022, in Berlin. The description can be found online¹³. About a dozen people participated, with most of them German-speaking. Bits und Bäume conference aims to bring together people interested or working in technology and digitalisation (what the conference refers to as the "bits" part) and sustainability and ecology (what they refer to as the "Bäume", German for trees). The first author provided physical copies of the card deck in both German and English. Several groups of two and one group of three then discussed for 20 minutes in-group, followed by 20 minutes' discussion in plenary. Two participants were researchers who work on accessibility of open source software.

4 CASE STUDIES FEEDBACK

In all case studies, participants reported that the card deck was an appropriate method to inform about knowledge on diversity and inclusion. The decision to include explanations about technical terms in footnotes on the cards rather than in a collection, produced positive feedback across the board. For example, one person at BuB reported that "it was really good to have the explanations on the card, I learned about intersectionality this way". This way, participants could gather information on new concepts on their own terms, rather than having to ask someone for an explanation, or go look for a leaflet somewhere, which could have meant an outing regarding their lack of knowledge. In some cases, participants read the explanation provided on their card to the rest of the group when in the discussion the meaning was unclear. The students in "Ways of Thinking" reported their primary takeaway to be the realization that even in rather current works, there were still opportunities to be more inclusive or sensitive towards marginalized stakeholders. More importantly, they also indicated that the language was appropriate, i.e. good to understand even without a background in either Gender Studies or HCI.

At both ditact and BuB, individual participants started out as rather skeptic after hearing about the concept of the card deck and looking at the card they had drawn, respectively. Both ended up very interested and reflecting positively about the card deck as a tool; the participant at BuB was taking extensive notes, nodding along while talking to their group partners (and observation not possible at ditact due to its online setting). This and the feedback on language described above show that the card deck can be used as a learning tool by people with different backgrounds.

On several occasions, participants described the card design as smart or interesting, especially regarding the temporal grouping of the recommendations. The prompt questions on the cards were considered by many to be positively thought-provoking. At both ditact and BuB, participants explicitly asked whether it was possible to obtain the card deck somewhere either as a printed set, or to print

themselves, so they could apply it after the respective workshops in their own environments.

At ditact, GE Academy and MCH, individual participants decided to exchange the cards they had originally drawn for another, as they considered them to not be applicable to their area. At BuB, one group asked for further explanations and additional examples especially for the "respect participants' identity" card. This indicates that some cards might have to be reframed or reworded to be more broadly applicable or easier to understand across different contexts.

Regarding timing, the first author had prepared schedules for the single-session workshops, so it would be easier to keep the time. The events ranged from 90 minutes (including one short break; MCH, BuB) to half-day workshops (including several breaks; ditact, GE Academy)¹⁴. In all cases the participants liked how the workshops were scheduled. In some settings, individual participants indicated a desire for further discussions on the recommendations themselves or issues of diversity and inclusion in general.

Throughout the different workshops, participants liked how different settings (individual and group settings) were used. At ditact, the participants described their mix of different backgrounds as fruitful for discussions. At GE Academy, the only additional feedback on the workshop was that some participants had hoped for more specific impulses and suggestions for their work. Giving specific advice, however, is (1) difficult to do without having in-depth knowledge of a project's context and (2) not what the first author had in mind with the cards anyways, so we see this as more of an issue regarding communication of the workshop contents.

Feedback from the MCH workshop delivered more self-reflection on workshop skills than explicit feedback on the card deck. The participants at MCH showed very different approaches toward diversity. The retiree who now works at a theater remarked that his team was very "old white ladies", and would like to diversify that, and appeared to be open to, and simultaneously struggling with, concepts of non-binary, mutable gender. Most of the other participants arrived at the workshop with the intent of talking about diversity in their hackspaces or workplaces, which were reported as "very white, cis guys of various ages", often related to their closeness to technology. The participants' differences in knowledge and understanding about (mostly) gender diversity led to engaged discussions, including some tension with the participants who had introduced themselves as non-binary or using non-gendered pronouns (e.g. they/them). While the participants remained respectful towards each other, this shows a necessity for workshop facilitators to have good skills in moderation, group facilitation and conflict management.

The workshop at Bits und Bäume was conducted in a closed room. Unfortunately, this made the small group exercise a bad experience for neurodivergent folks, because the room acoustics were rather poor.

5 DISCUSSION, LIMITATIONS, FUTURE WORK

As the authors had planned for/assumed beforehand, using the cards works well in various group sizes. People can engage with single cards individually, or can discuss a selection of cards in

¹²<https://bits-und-baeume.org/en/>

¹³<https://fahrplan22.bits-und-baeume.org/bitsundbaeume/talk/NTYXZH/>, last accessed 2023-03-04

¹⁴This difference in online vs. offline settings is owed to the structure of the events; neither Bits und Bäume nor MCH allow for half-day workshops.

small groups, and the findings of these engagements can then be discussed in the plenary. A thing that has not worked so well was documenting the findings together - two ways to tackle this could be either audio recordings, or appointing a designated note-taker (not a participant, ideally). Overall, the participants considered the cards a good tool. In several workshops, small issues came up with the examples on the cards: some need to be reworked to reflect the goal of being multi-dimensional (i.e., where possible, add examples and explanations that do not relate to gender, but e.g. to race or disability), and some texts could be rephrased to be more accessible.

The insights from the workshops show that it is important to have good workshop facilitation skills to keep group work going and make it a good experience for everyone, as topics of marginalization, oppression, and related advocacy work are sensitive topics. This includes having established a good understanding of the recommendations and the underlying concepts as well as a certain amount of leadership, or rather, educators' skills.

Most participants in all workshops had not known each other beforehand, which only allows for limited observations regarding ad-hoc power structures (as compared to the established structures one would have to observe and work with in, e.g., research groups). The ad-hoc character of the workshop groups might have been the reason why no specific/hands-on next steps were discussed. Aside from the meta-goal of "making our surroundings more inclusive", the participants did not share a specific goal or issue. This is something to plan for in future workshops: go into more specifics, see what the participants themselves can come up with. This would further entail deeper dives into their projects, or to prepare a scenario to work through together.

However, for now, the workshops described in this paper have shown that the Anti-Bias Card Deck can be used in different group settings for participants with differing prior expertise on issues of diversity and inclusion. Insights from the workshops can inform further improvements to the card deck itself as well as to future workshops using them.

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